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DOCUMENT TITLE:

PRELIMINARY SPECIFICATION OF TFT MODULE TYPE

CUSTOMER	DYSON
CUSTOMER PART NO.	**X692**
MODEL NUMBER	**COG-VLUK7090-01**
CUSTOMER APPROVAL	Thanarajan
DATE	13/06/24

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CONTENTS

BOE VARITRONIX LIMITED

Preliminary Specification
Of
TFT Module Type
Model No.: COG-VLUK7090-01

1. General Description

1.1 Introduction

This is a 1.44-inch TFT Module. TFT-LCD Panel and backlight system.

1.2 Application

Industrial

1.3 General Specifications

Table 1: General Specification

Parameters	Specifications	Unit
Display Size (Diagonal)	1.44"	inch
Aspect Ratio	1:1 (landscape)	-
TFT-LCD Controller	ST7735P3-G6	
Number of Pixels (H x V)	128 x BGR x 128	-
Color Configuration	BGR vertical stripes	
PPI	125	pixels
Pixels Shape	Square	-
LCD Type	ADS, transmissive, amorphous silicon TFT	-
Color Resolution	262K colors	-
Brightness(on panel surface)	Typ. 350	cd/m^2
Image Mode	Normally Black	-
Contrast Ratio @Ta = 25°C	Typ. 800: 1 (Perpendicular viewing)	-
Viewing Angle (12/6/9/3 o’clock) (Note 1)	All directions	degree
Operating Temperature	-20 to +70	°C
Storage Temperature	-30 to +80	°C

Regulation	RoHS compliant	-
Electronics and Interface		
TFT Display Interface Method	• SPI interface • 12-pin FPC	-
Basic Display Features (TFT)		
Visibility with Polarized Sunglass	Visible at landscape	-
Display Power Consumption	Typ. 33	mW
Backlight Power Consumption	Typ. 54	mW

Note:
1. At the U/D/L/R direction, the viewing angle is same;
2. FPC layout (GM1 to GND, GM0 to VDD)

GM1, GM0	I	-Panel Resolution Selection Pins.	VDDI/DGND

GM1	GM0	Selection of panel resolution
0	0	132RGB x 162 (S1~S396 & G1~G162 output)
0	1	132RGB x 132 (S1~S396 & G1~G132 Output)
1	1	128RGB x 160 (S7~S390 & G2~G161 output)

2. Mechanical Specifications

The mechanical detail is shown in Fig. 1 and summarized in Table 2 below.

Table 2: Module mechanical detail

Parameters		Specifications	Unit
Outline dimensions		46.6(W) x 34.16(H) x 5.96(D) (Exclude FPC, cables, post & component)	mm
Color TFT 128 x BGR x 128	Active area	25.498(W) x 26.50(H)	mm
	Sub-Pixel pitch	0.0664(*3)(W) x 0.207(H)	mm
Surface	Surface treatment	Polarizer with AG treatment	-
	Surface hardness	≥2	H
Backlight		White LED	-
Weight		TBD	gram

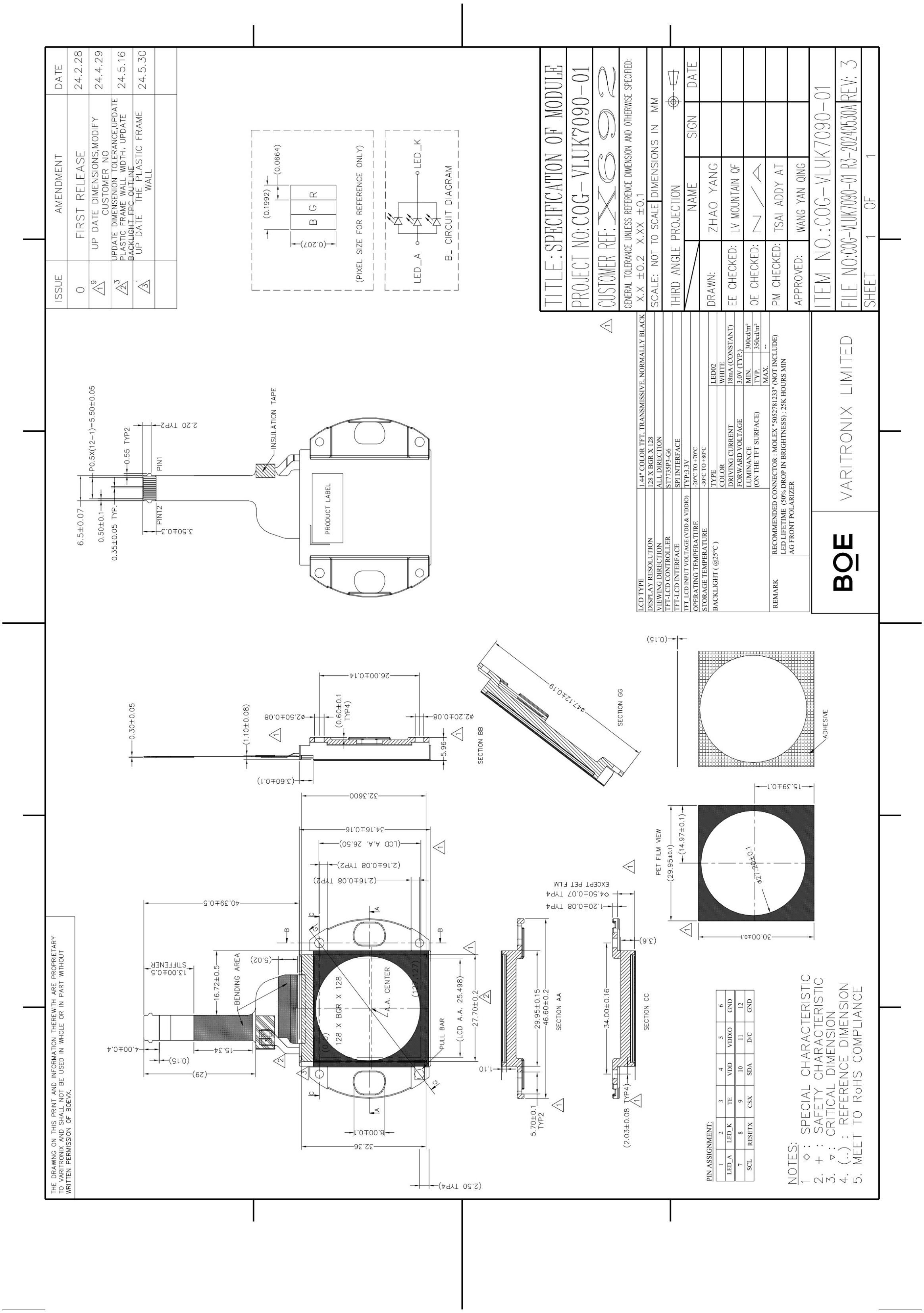


Figure 1: Module specification

3. Interface Signals

3.1 TFT-LCD Panel Driving

Recommend connector item(CON): MOLEX “5052781233” (NOT INCLUDE)

Table 3: Connector Pin Assignments for TFT

Pin No.	Symbol	I/O	Description	Remark
1	LED_A	I	LED Anode	
2	LED_K	I	LED Cathode	
3	TE	O	Tearing effect output pin to synchronies MCU to frame rate, activated by S/W command.	If not used, please open this pin
4	VDD	P	Power pin for Logic	
5	VDDIO	P	VDDIO Voltage Output Level for Monitoring	
6	GND	P	Ground	
7	SCL	I	Display data/command Selection Pin in MCU Interface	D/CX=’1’: Display Data or Parameter. D/CX=’0’: Command Data.
8	RESETX	I	This signal will reset the device and it must be applied to properly initialize the chip.	Signal is active low.
9	CSX	I	Chip Selection Pin	Low Enable.
10	SDA	I/O	SDA is the serial input/output signal in serial interface mode.	-
11	D/C	I	In 4-line SPI, this pin is used as D/CX (data/ command selection).	If not used, please fix this pin at VDDIO or DGND level.
12	GND	P	Ground	

Remark: For I/O, “I” is input, “O” is output, “P” is power; “C” is passive

Note 1: The orientation of the module, please refer to module drawing.

4. Absolute Maximum Ratings

4.1 Electrical Maximum Ratings

The product or its functions may subject to permanent damage if it's stressed beyond those absolute maximum ratings listed below. Exposure to absolute maximum rating conditions for extended periods may affect display module reliability.

At GND=0V

Table 4: Absolute Maximum Ratings & Environmental Conditions

Item	Symbol	Min.	Max.	Unit
TFT-LCD supply voltage (analog)	VDD	-0.3	+3.96	V
TFT supply voltage for Logic	VDDIO	-0.3	+3.96	V
TFT digital I/O input signal	V_{IO}	-0.3	VDDIO+0.3	V
Single LED forward current	IF	-	18	mA
Relative Humidity (at 60°C) (Note 2)	RH		90	%
Operating Temperature (Note 1,2)	Topr	-20	+70	°C
Storage Temperature	Tstg	-30	+80	°C

Note 1: Panel surface temperature should not exceed 80°C.

Note 2: No condensation allowed under any condition.

[Caution]

Do not display fixed pattern for prolonged hours because it may develop image sticking on the display.

5. Electrical Specifications

5.1 Block Diagram

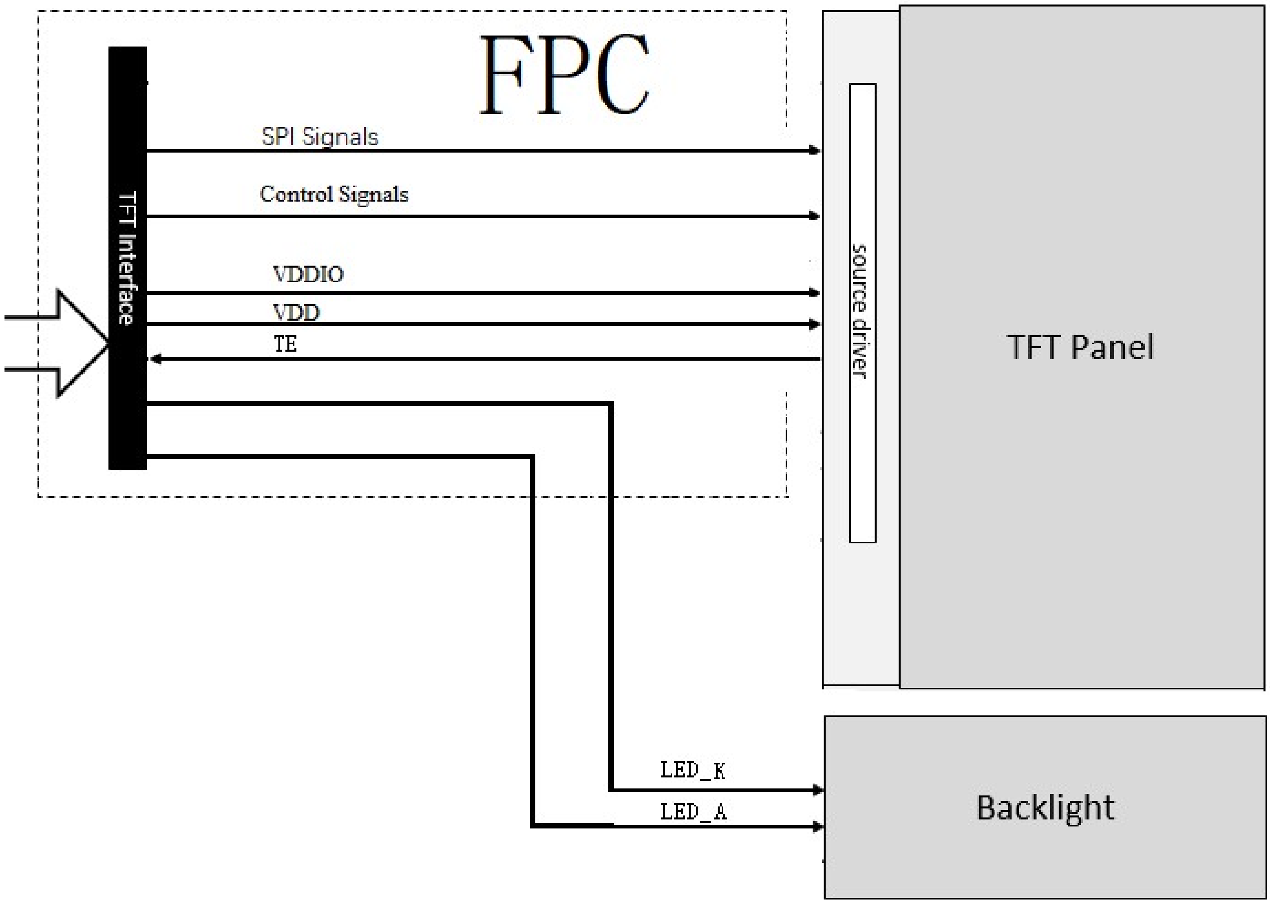


Figure 2: Block diagram

Remark: signals applied pins

1. Control Signals: RESETX,
2. SPI Signals: CSX, SCL, SDA, D/C,

5.2 Typical Electrical Characteristics for TFT-LCD

At Ta = 25 °C, VDD=3.3V, GND=0V

Table 5: DC Characteristics for TFT LCD

Parameter	Symbol	Min.	Typ.	Max.	Unit
TFT-LCD supply voltage (analog)	VDD (Note 1)	3.15	3.3	3.45	V
Power supply voltage for Logic	VDDIO(Note 1)	3.15	3.3	3.45	V
TFT-LCD supply current (analog)	IDD (Note 2)	-	5	-	mA
Power supply current for Logic	IDDIO (Note 2)	-	5	-	mA
Input signal high voltage	V_{IH} (Note 3)	0.7*VDDIO	-	VDDIO	V
Input signal low voltage	V_{IL} (Note 3)	GND	-	0.3*VDDIO	V
Output signal high voltage	V_{OH} (Note 4)	0.8*VDDIO	-	VDDIO	V
Output signal low voltage	V_{OL}(Note 5)	GND	-	0.2*VDDIO	V

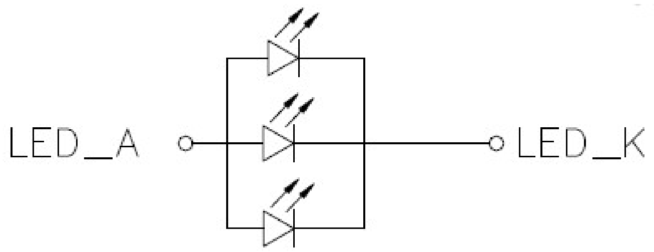
Note 1: The supply voltage is measured and specified at the interface connector of the module.
Note 2: Tested at all white pattern.
Note 3: Applied pins: all input and output pins with VDD.
Note 4: IOH = -1.0mA
Note 5: IOL = +1.0mA

5.3 Recommended Driving Condition for LED Backlight

Table 6: LED Driving specifications (Ta = 25°C)

Parameter	Symbol	Condition	Min.	Typ.	Max.	Unit	Remark
Supply voltage of LED backlight	V_{LED}	Backlight current = 18mA $I_{LEDTotal}$ (Typ.)	-	3	-	V	Note 1
Supply current of LED backlight (1 String)	I_{LED}	Per LED string	-	6	-	mA	Note 2
Total Supply current of LED backlight	$I_{LEDTotal}$	$I_{LED1} + \ldots + I_{LEDn}$ (n=LED string numbers as Note 1)	-	18	-	mA	Note 2
Backlight Power Consumption	P_{LED}	-	-	54	-	mW	Note 3
LED life time	LIFE	Ta=25°C, I_{LED}=Typ. value	25000	-	-	Hrs	Note 4

Note 1: Backlight Circuit Diagram



Note 2: Each string LED should be drove by constant current separately.
Note 3: Backlight power consumption is calculated by I_{LED} (Total) x V_{LED}
Note 4: The LED Life-time define as the estimated time to 50% degradation of initial luminous.
Note 5: Recommended derating curve (current per LED string) for backlight driving

TBD

Figure 3: LED driving duty derating curve

5.4 TFT-LCD Timing Characteristics

5.4.1 Serial Interface Characteristics (4-line Serial)

At Ta = 25 °C, VDD=3.3V, GND=0V

Table 7

Signal	Symbol	Parameter	MIN	MAX	Unit	Description
CSX	TCSS	Chip Select Setup Time (Write)	45		ns	
	TCSH	Chip Select Hold Time (Write)	45		ns	
	TCSS	Chip Select Setup Time (Read)	60		ns	
	TSCC	Chip Select Hold Time (Read)	65		ns	
	TCHW	Chip Select “H” Pulse Width	40		ns	
SCL	TSCYCW	Serial Clock Cycle (Write)	20		ns	-Write Command & Data Ram
	TSHW	SCL “H” Pulse Width (Write)	10		ns	
	TSLW	SCL “L” Pulse Width (Write)	10		ns	
	TSCYCR	Serial Clock Cycle (Read)	150		ns	-Read Command & Data Ram
	TSHR	SCL “H” Pulse Width (Read)	60		ns	
	TSLR	SCL “L” Pulse Width (Read)	60		ns	
D/CX	TDCS	D/CX Setup Time	10		ns	
	TDCH	D/CX Hold Time	10		ns	
SDA (DIN) (DOUT)	TSDS	Data Setup Time	10		ns	For Maximum CL=30pF For Minimum CL=8pF
	TSDH	Data Hold Time	10		ns	
	TACC	Access Time	10	50	ns	
	TOH	Output Disable Time	15	50	ns	

Note: The rising time and falling time (Tr, Tf) of input signal are specified at 15 ns or less.
Logic high and low levels are specified as 30% and 70% of VDDI for Input signals.

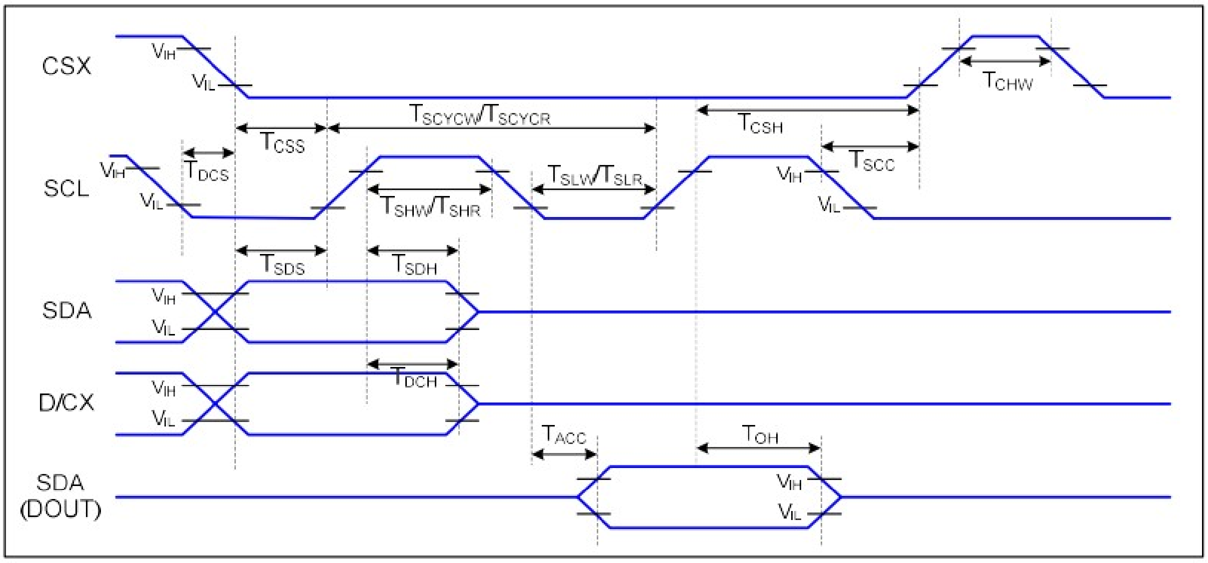


Figure 4: 4-line Serial Interface Timing

5.5 Reset Timing Characteristics

At Ta = 25 °C, VDD=3.3V, GND=0V

Table 8

Related Pins	Symbol	Parameter	MIN	MAX	Unit
RESX	tRESW	Reset Pulse Duration	10	-	us
	tREST	Reset Cancel	-	5 (Note1,5)	ms
				120 (Note1,6,7)	ms

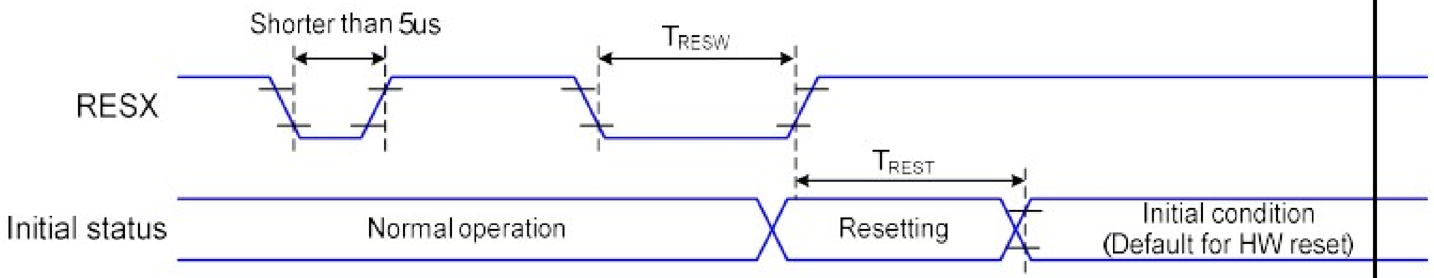


Figure 5: Reset Timing Chart

Notes:

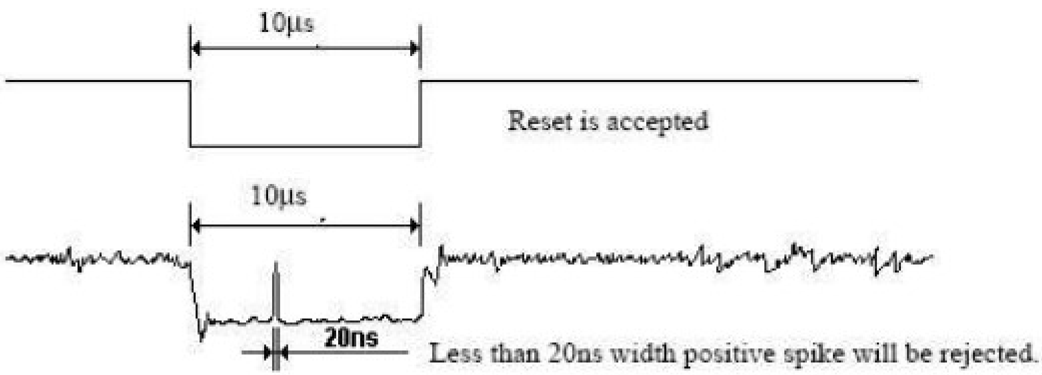
1. The reset cancel includes also required time for loading ID bytes, VCOM setting and other settings from NVM (or similar device) to registers. This loading is done every time when there is HW reset cancel time (tRT) within 5 ms after a rising edge of RESX.

2. Spike due to an electrostatic discharge on RESX line does not cause irregular system reset according to the table below:

RESX Pulse	Action
Shorter than 5us	Reset Rejected
Longer than 9us	Reset
Between 5us and 9us	Reset Starts

3. During the Resetting period, the display will be blanked (The display is entering blanking sequence, which maximum time is 120 ms, when Reset Starts in Sleep Out –mode. The display remains the blank state in Sleep In –mode) and then return to Default condition for Hardware Reset.

4. Spike Rejection also applies during a valid reset pulse as shown below:



5. When Reset applied during Sleep In Mode.

6. When Reset applied during Sleep Out Mode.

7. It is necessary to wait 5msec after releasing RESX before sending commands. Also Sleep Out command cannot be sent for 120msec.

5.6 Power On/Off Sequence

The power on/off sequence is illustrated below

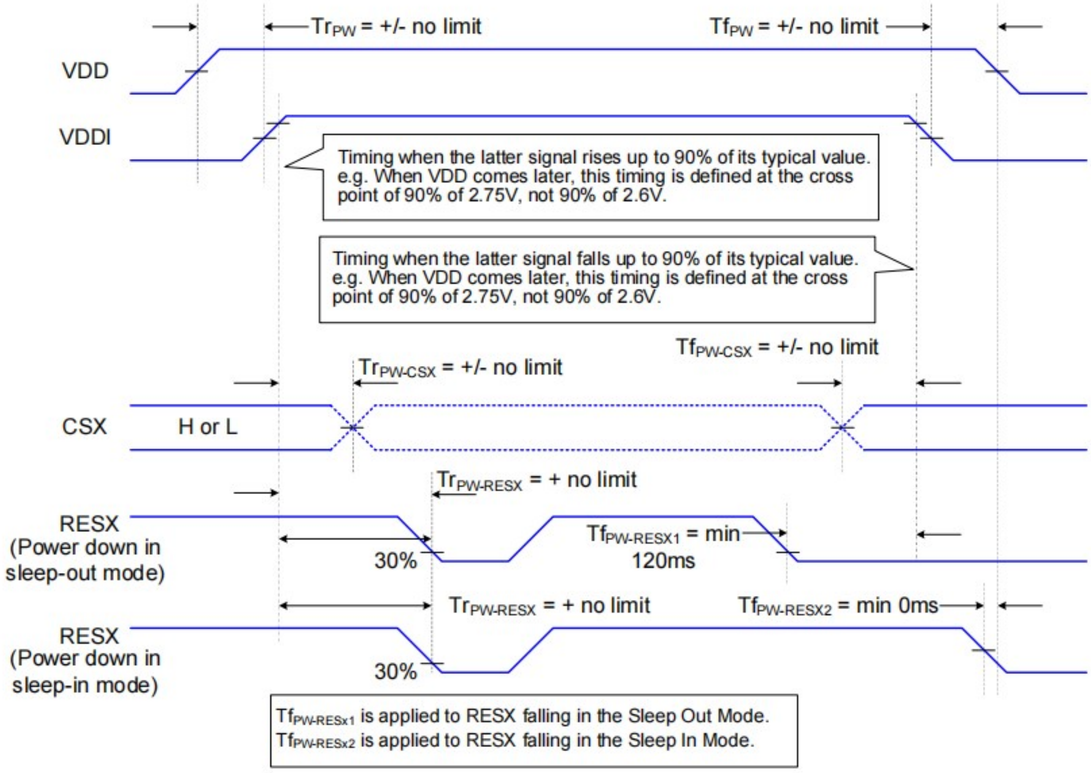


Figure 6: Power On/Off Sequence

5.7 Initial code setting (for reference only)

Table 9

TBD

6. Optical Characteristics

Conditions unless specified otherwise:

- TFT-LCD supply voltage = 3.3 volts
- Elapsed time from switch on is greater than 30 minutes
- RGB, white and black test patterns only
- Factory settings
- Luminance = 100% unless specified
- Measurements are conducted at ambient temperature and perpendicular on module center unless specified

Table 10: Optical Characteristics

Parameter		Symbol	Condition	Min.	Typ.	Max.	Unit	Remark
Contrast Ratio		CR	Viewing angle θ=ϕ=0°	500	800	-		Note 1
Luminance(on the cover lens surface)		L	Viewing angle θ=ϕ=0°	300	350	-	cd/m^2	Note 1
Viewing angle ($C_r \geq 10$)			-	75	85	-		Note 1
				75	85	-		
				75	85	-		
				75	85	-		
Uniformity White		Ub		75	-	-	%	Note 2
Chromaticity	Red	x	Viewing angle θ=ϕ=0°	TBD	TBD	TBD	-	Note 3
		y		TBD	TBD	TBD	-	
	Green	x		TBD	TBD	TBD	-	
		y		TBD	TBD	TBD	-	
	Blue	x		TBD	TBD	TBD	-	
		y		TBD	TBD	TBD	-	
	White	x		TBD	TBD	TBD	-	
		y		TBD	TBD	TBD	-	
NTSC		NTSC		40	50	-	%	

Note 1: View angle and Contrast、Luminance definition:
Contrast shall be measured at the center of the LCD surface by using DMS or similar equipment.
Luminance shall be measured with all pixels set to full white, then to full black state.

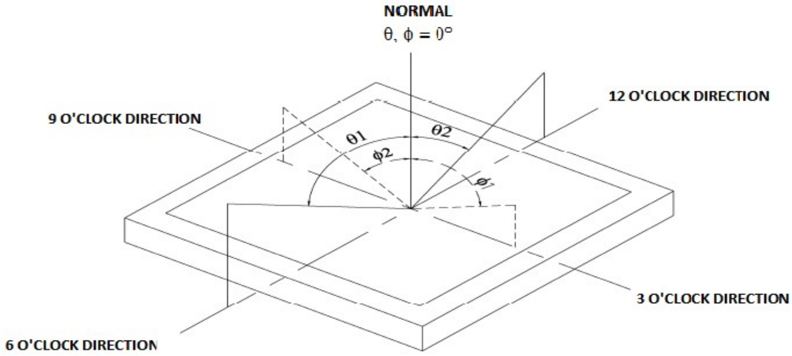


Figure 7: Viewing Definition

Contrast Ratio (CR) is defined mathematically.

$$CR = \frac{\text{Luminance when displaying a white raster}}{\text{Luminance when displaying a black raster}}$$

Note 2: Measurement method according to 5 points method.

$$\text{Uniformity } \Delta Y = \frac{\text{Minimum Luminance of 5 points}}{\text{Maximum Luminance of 5 points}} \times 100\ (\%)$$

Figure 8: Transmittance uniformity test method

Noet.3: The color chromaticity coordinates specified is reference to actual spectral data measured with all pixels first in red, green, blue and white. Measurements were made at the center of the panel.

7. Reliability Tests / Environmental

7.1 Reliability Test Conditions

Table 11: List of reliability tests

Test		Symbol	Condition	Reference	Quantity
1	High Temperature Storage	HST	+80°C / 72 hrs	IEC 60068-2-2 Bb	4 PCS
2	Low Temperature Storage	LST	-30°C / 72 hrs	IEC 60068-2-1 Ab	4 PCS
3	High Temperature Operating	HOT	+70°C / 72 hrs	IEC 60068-2-2 Bb	4 PCS
4	Low Temperature Operating	LOT	-20°C / 72 hrs	IEC 60068-2-1 Ab	4 PCS
5	High Temperature High Humidity Test Operating	HTHH	+60°C / 90% RH / 72 hrs	IEC60068-2-78 Cab	4 PCS
6	Temperature Shock Test	TST	-20°C <> +70°C 30min/5min/30min, 10cycles Non-operating	IEC 60068-2-14Na	4 PCS

Notes:

1. The test result shall be evaluated after the sample has been left at room temperature and humidity for at least 2 hours without load.
3. After the reliability test, the product only guarantee function normally without any fatal defects (no display, line defect, abnormal display etc.).
4. After completion of the test, when there is no agreement with customer, the sample should be free from the following defects:
   a) Air bubbles in the LCD
   b) Seal leakage
   c) No display
   d) Missing lines
   e) glass crack
   f) Current consumption increment > 100%
   g) Contrast reduction > 40%
   h) Luminance reduction > 40%
   i) Color coordinates change > 0.05
5. Performs backlight de-rating during high temperature operating test
6. Module color change (such as yellowish) will be judged as pass if the color coordinates variation is less than or equal to 0.05.
7. No sequential tests are guarantee unless the supplier has agreement with customer.
8. Optical performance below -20°C is not guarantee.

7.2 Electrostatic Discharge (ESD)

Table 12: ESD test conditions

Test	Condition	Method	Remark	Quantity
Human body model	R = 330Ω, C = 150pF, • Air discharge: ±10 KV to display surface • Contact discharge: ±6 KV to display surface	IEC61000-4-2	Not operating	2pcs

Note 1: The TFT-LCD panel and IC on module are sensitive to electrostatic discharge; please make sure equipments and operators are properly ground before and during handling.

Note 2: As different customer application have different interfacing designs and assembly processes, the display module has no ESD protection circuitry. Customer is required to take special care on ESD level control in the assembly and test processes.

8. Inspection Criteria

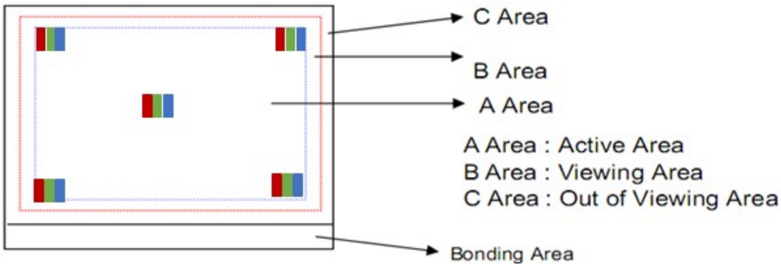
8.1 Inspection Conditions 检查条件

Items 项目	Condition 条件	
Ambient lighting 环境亮度	Cosmetic inspection 500~1000 Lux. function inspection less than 200 Lux. 外观檢查 500~1000 Lux. 功能檢查小于 200 Lux	
Temperature /Humidity 温度/湿度	21 ± 4°C with 35～80%	
Driving condition 驱动条件	Equipment 设备	Product specific test tool 产品规格书说明的测试工具
	Test pattern 测试画面	Black, White, Red, Green, Blue, etc 黑，白，紅，綠，藍，等
	Supply voltage 电源	Typical voltages as given in the specification 规格书提供的典型电压
Inspection method 检查方法	Distance 距离	40 cm ± 10 cm from display 距显示屏40cm ± 10 cm
	Viewing angle 视角	performing in front of the panel All directions for inspecting the sample should be: ADS type： within 45º to perpendicular line.； TN type： within 10º to perpendicular line.； ADS 产品：面向 Panel，所有方向 45°角内（与垂直线）； TN 产品：面向 Panel，所有方向 10°角内（与垂直线）；

8.2 Visual Inspection Criteria

Items 项目	Details 详细检查点		Inspection Criteria 检验标准		Type
			A Area	B/C Area	
Dot Defects function defect) 亚像素缺陷(功能不良)	Bright dot Defect 亮亚像素缺陷		N≤0	Ignore	Major
	Dark dot Defect 暗亚像素缺陷		N≤2		
	Bright + Dark dot Defect 亮+暗亚像素缺陷		N≤2		
	Joint dot Defect(Combining with bright dot standard to judge) 亚像素相连缺陷(结合亮点标准判定)		N≤1		
Line Defects 线缺陷	Bright Line, Dark Line 亮线,暗线（像素不良导致）		N=0		
Cosmetic defect 外观不良	Foreign material Black/Bright Spot 异物暗点亮点 (Hair, Lint, etc) (头发，棉绒等)	Circular Type 圆型	Size（大小） Accepted qty（接受数量）: D≤0.2 Ignore; 0.2m<D≤0.3mm, N≤2, D>0.3mm, N=0	Ignore	inor
		Linear Type 线型	W≤0.05mm, Ignore 0.05mm<W≤0.1mm, L≤3mm, N≤2 W>0.1mm, L>3mm, N=0		
	Galaxy dot 满天星	Visible under 5%ND at black pattern 5%ND 可见碎亮点（碎亮点0.2mm以下）全黑画面 Invisible under 5%ND at black pattern 5%ND 不可见碎亮点（包括碎亮点0.2mm以下）全黑画面	N≤3 inside the area of D=15mm 15mm 直径范围内，N≤3 Ignore 不计		
		Consecutive galaxy dot along a line at black pattern 连续成线性碎亮点 全黑画面	N=0		
	dent or bubble 凹痕或气泡	Circular Type 圆型	D≤0.2, Ignore 0.2mm<D≤0.3mm, N≤2 D>0.3mm, N=0		
	Scratch 划痕	Linear Type 线型	W≤0.05mm, Ignore 0.05mm<W≤0.1mm, L≤3mm, N≤2 W>0.1mm, L>3mm, N=0		
	Abnormal display 异常显示	All pattern 全画面	Not allowed 不允许	Ignore	Major
	Mura, 不均匀 (Pure black / white & 50% grey pattern check 纯黑/白&50%灰画面检查)		Refer to limit sample agreed by both BOEVX and customer or 5%ND Filter, check time less than three seconds, the distance from ND filter to product around 5cm,from ND filter to eye around 30cm~40cm, checking time:3s 参考双方承认极限样品或者在 5% ND 滤片下检查，检查时间<3s。ND filter 距离产品约 5cm, 距离眼睛约 30cm~40cm	Ignore	Minor

Note 1) Definition of the Area 区域定义



Active Area:有效区域 Out of Viewing Area: 可视区之外

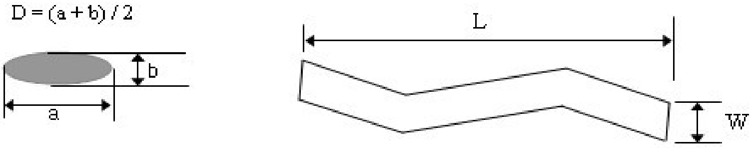
Viewing Area:可视区域 Bonding Area: 邦定区域

Note 2) Protective film scratch/bubble/ surface dirt are not accounted.保护膜划伤/保护膜气泡/表面脏污不计不良

Note 3) D = Diameter (直径), L = Length(长度), W = Width (宽度), N = Number (数量), DS = Distance.

Spot which caused by foreign material(particle) will not be counted the color or the brightness. Spot size will be counted by the actual measurement of the particle. If one particle judged by D or (L,W) is within spec, then this particle will be accepted.

异物暗、异物亮点、斑点等非像素点类缺陷，不计颜色和亮度。 大小按异物实际测量计算；若按圆型（D）或线型之一(L、W)的标准判定可接受，则此点缺陷可接受。



Note 4) [1] 1pixel =3dots(Dots：sub-pixels 即亚像素). Dot which is invisible through 5%ND filter or smaller than 1/2 of sub-pixel size will not be counted as "1 dot"defect.

5%ND可遮盖或者大小小于1/2亚像素的点不计算为“1点”缺陷.如下图：

"No dot defect"	"1 dot defect"	"1 dot defect"	" No dot defect "
“没有不良点”	“1 个不良点”	“1 个不良点”	“没有不良点”
(=ignored/not counted)	(=counted)	(=counted)	(=ignored/not)

Bright Dot Defects: Dots(sub-pixels) on display which appear bright in the display area at R/G/B.

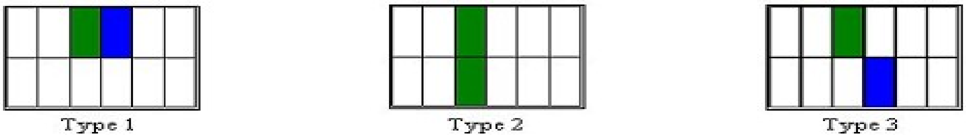
显示区域显示时在R/G/B画面下可见的亮的像素点（sub-pixels）

Dark Dot Defects: Dots(sub-pixels) on display which appear dark in the display area at R,G,B Color Pattern.

显示区域显示时在R/G/B画面下可见的暗的像素点（sub-pixels）

[2] adjacent dots defect]

两个相邻亮子像素缺陷

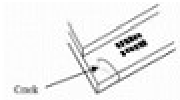
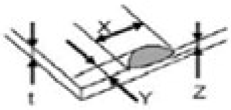
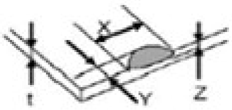
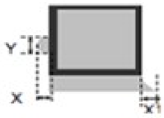


Note 5）Mura: Mura on display which appears darker / brighter against background brightness on parts of display area at L0/L127/L255

显示区域显示时在 L0/L127/L255 画面可见的发亮程度不均一的现象区域

8.3 Appearance Inspection Criteria

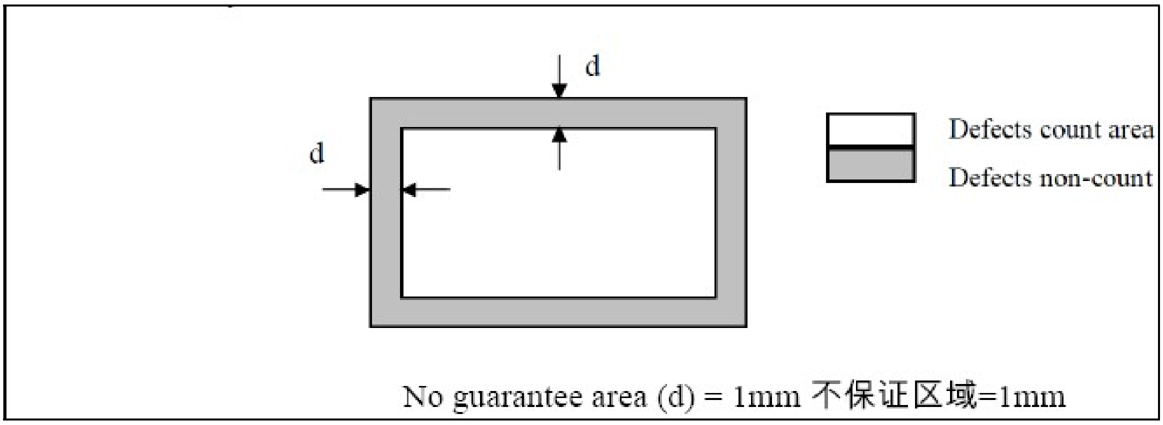
8.3.1 TFT Panel defects

Defects 缺陷		Criteria 标准	Type 类型
Crack 裂纹		Not Allowed 不允许	Major 重要
CF Side Chipping CF 边崩缺		Y≤1/2 BM X Ignore 忽略 Z≤t	Minor 次要
TFT Side Chipping TFT 边崩缺		Y≤1mm X Ignore 忽略 Z≤t	Minor 次要
Corner Chipping 边角崩缺		X+Y≤4mm Z≤t	Minor 次要
Burr 毛刺		X+X1≤0.2mm 不影响功能、画面及组装	Minor 次要

Remark:

1. *The above shall not affect the function and assembly 以上均不得影响线路功能及组装
2. * Surface dirt and gelling can be wiped is OK.表面脏污及黏着可擦拭为 OK
3. * The chipping cannot reach to the perimeter. 崩缺不能进入 TFT 框胶

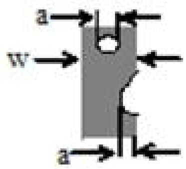
8.3.2 Glue Extending and shortage defect



Note: Definition of defects counting: we inspect defects among film, release film, protective film, and do not consider the non-count area (no progressive defect is allowed)

备注：缺陷计数的定义：我们检查薄膜、释放膜、保护膜之间的缺陷，不考虑非计数区域(不允许有进行性缺陷)

8.3.3 FPC defects, FPC 缺陷

Items 项目	Description 描述	Acceptance criteria 验收标准
FPC pad 	Dent, Pinhole, Incomplete, 线路凹痕，针孔，不完整	Acceptable if a ≤ w/3 如果a≤ w/3可接受
	Fold mark 线路折痕	Acceptable if the circuit is not broken 如果金手指线路没断开可接受
	Open circuit 金手指开路	Not Acceptable 不能接受
	Oxidation, Inerasable contamination 金手指氧化、擦不掉的玷污	Not Acceptable 不能接受
FPC fold mark FPC 折痕	without metal circuit damage 不导致线路损坏	No count 不计数

8.3.4 Other defect

Items 项目	Criteria of acceptance 验收模式	Inspection pattern 检查模式
Light leakage 漏光	Not visible in 30° viewing cone。 30° 视锥内检查不出来不可见。	Pure white/Pure black 纯白/纯黑
Newton's rings 牛顿环	D≤5mm, ignore; D>5mm, not allow. D≤5mm,不计；D>5mm,不允许。	Use general cosmetic inspection condition 使用外观检查条件进行检查
Backlight BZ/ metal BZ scratch（No surface treatment） 背光Bezel/Metal Bezel划伤（无表面处理） 	Finger touch and no touch feeling 裸手指触摸无触感。 （此检查方法有安全风险）	Finger touch 裸手触摸
BZ scratch standard (with surface treatment) BZ划伤(有表面处理类)	The depth is no into the base material is accepted. For those electrophoretic paint, repaired is accepted. The accepted repaired point is ≤ 5. 深度不入基材可接受，电泳漆表面有划痕或者刮花补油后可接受，可接受的修补油漆数量为5个	Visual inspection 目视
EMI tape delamination and damage EMI胶纸翘起及破损	Delamination angle should not exceed 90° and tape should not curl. Length of delamination or damage should not exceed 3mm. Acceptable if assembly process is not affected 边角翘起在90° 范围内，不可卷曲，翘起/破损长度不超3MM，不影响功能和装配可接受	Visual inspection, caliper 目视，卡尺
Ink Printing/ label printing 喷码	Key information can be recognized will be accepted 关键信息可识别即可接受	Visual inspection 目视
BSR (buzz, squak, rattle noise) and odor 异响及气味	BSR and odor according to BOE internal control standard 异响及气味标准根据内部标准管控	NA

8.4 BOE 客户质量服务流程 （BOE Customer Quality Service Process）

In order to provide better service to Customer, BOE shall apply the after-sales product quality service process as below
为了提供给客户更好的服务，BOE 应该提供如下的售后产品质量服务过程：

8.4.1、According to the P/O from Customer, BOE should deliver required product to the place appointed by Customer. 根据P/O，BOE将产品运送到客户指定地点.

8.4.2、Customer will do IQC for the incoming product. 客户要对来料产品做IQC检查。

8.4.3、Inspection standard should be provided by BOE, and it will be valid after confirmed by Customer. Inspection and Defects determination should be carried out according to the standard agreed by both Parties.检查基准由BOE提供，并由客户确认通过，检查手法和不良按双方达成的基准协议确认。

8.4.4、Customer should use the LCD product according to the instruction. BOE will not be responsible for the defect product caused by violation of Users' Instruction.
客户使用BOE产品要遵守说明书，对于违反说明书的使用BOE不负责。

8.4.5、Both parties should deal with the quality problem with friendly cooperative policy. And both parties should negotiate to deal with the defect products of which the responsibility is not very clear 双方在处理产品质量问题时要遵循友好合作策略，对于责任方归属不明确时，双方要谈判解决。

8.4.6、If there is no other signed document, the warranty of the product is 12 months after the production date (the date on the product label). 如无其他签署协议，产品保质期为12个月，从生产日期（即产品label日期）开始算。

8.5 The warranty will be avoided 质保期失效:

The warranty will be avoided in cases of below: 在下列情况中，质保期将会失效：

8.5.1、When the warranty period is expired. 质保时间超期。

8.5.2、When the LCMs were repaired by 3rd party without Supplier's approval.
LCM在没有供应商的允许下交由第三方修复。

8.5.3、When the LCMs were treated like disassemble and rework by the Customer and/or customer's representatives without Supplier's approval. LCM在没有供应商的允许下，被客户或是客户的代表方拆开或是维修时。

8.6 Service scope on warranty part which confirmed by BOE its responsibility:

质保期内经BOE确认存在质量问题的产品，可提供如下服务：

8.6.1.、Base on customer request, provide service of repair, or 1to1 replace, or 1to1 credit at its selling price.

"三包服务"：根据客户需求对不良品进行无偿维修、或1对1退换货、或1对1按产品销售单价退款。

8.6.2、Base on customer request, BOE could provide the analysis and improvement report. After BOEVX sending out the report 15 working dates, if there is no further comment from customer, the issue will be closed.

根据客户需求提供相关分析改善报告。 BOE提供分析改善报告15个工作日后，若未收到客户进一步反馈，则问题关闭。

8.6.3、At any time, if third party or other related service is required for solving quality issue, both BOE and customer should have discussion and reach agreement in advance. Otherwise, BOE have the right to reject any extra cost without it authorization.

任何时候，若因品质问题需BOE提供第三方或者其他相关服务时，双方应提前沟通并协商一致。否则，BOE将拒绝任何未经BOE授权的额外费用。

Remark: Out of warranty period, no further service would be provided including repair, or replace, or credit service etc.

*备注：对质保期失效产品，BOE不再做相关的质保承诺，且不再提供"三包服务"或者其他相关服务。

8.7 Caution For operation 操作注意事项

8.7.1、Do not display the fixed pattern for a long time because it may develop image sticking due to the LCM structure. If the screen is displayed with fixed pattern, use a screen saver

由于LCM结构原因，不能长时间显示一个固定模式，会造成残像；如果屏幕显示为多种模式，请添加一个屏幕保护程序。

8.7.2、Static electricity (ESD) will damage the panel,. Please make sure that operators wear static-protective glove effectively and working tables &device are effectively grounded during operation and other ESD protective method

ESD会损坏Panel，确保作业员在作业过程中佩戴防静电手套，并且在作业过程中工作台和设备要有效接地。

8.7.3、Please place LCM on the tray provided by BOE while moving it, in order to avoid mechanical damage.

将LCM放在BOE Tray盘中转运，以防止机械损伤。

8.7.4、Please keep the LCM in the specified, original packing boxes when storage.
不能将没包装材料覆盖的LCM进行堆积重叠。

8.7.5、DO NOT press the area covered with PET or such materials. These are weak point of LCM since of TCPs (Driver ICs) and PWBs.
不要直接用手触摸TCP（驱动IC）、PWB板。

8.7.6、Since the LCM is made of glass, do not apply strong mechanical impact or static load onto it. Handling with care since shock, vibration, and careless handling may seriously affect the product. If it falls from a high place or receives a strong shock, the glass may be broken.
LCM的Panel包括两张薄玻璃，非常容易被损坏，所以在处理LCM时应该极其小心。LCM是由玻璃制造而成，因此表面不能承受住强烈的机械撞击或是静态的压力，在处理时避免撞击、振动。粗心对待会严重影响产品，如果LCM从高处掉落或是受到强烈的撞击，玻璃可能会碎掉。

8.7.7、The polarizers on the surface of panel are made from organic substances. Be very careful for chemicals that not to touch the polarizers or it may leads the polarizers to be deteriorated.
Panel表面的偏光片是由有机物构成，所以要避免化学品接触到偏光片，否则会导致偏光片的损坏。

8.7.8、Remove the protective film slowly, keeping the removing direction approximate 30-degree not vertical from panel surface, if possible, under ESD control device like ion blower, and the humidity of working room should be kept over 50%RH to reduce the risk of static charge.
撕除LCM保护膜要慢，角度大约为30°，不要垂直于Panel表面撕膜；可能的话在离子风机下，湿度50%下进行以降低静电风险。

8.7.9、Avoid the use work clothing made of synthetic fibers. We recommend cotton clothing or other conductivity-treated fibers.
工作布条应避免使用合成纤维，应使用棉布或是导电纤维布。

8.7.10、Avoid any force bending on FPC golden finger area.
FPC 金手指区域避免受力弯折

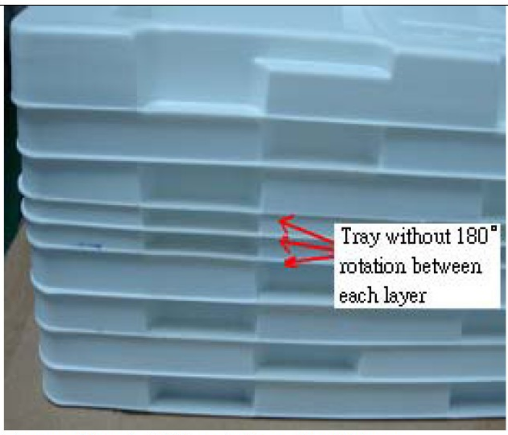
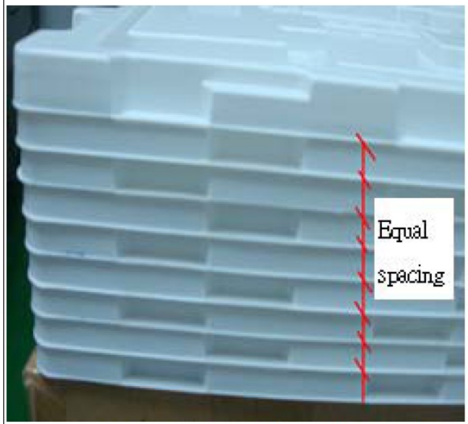
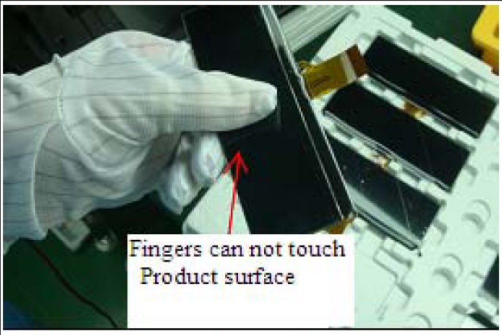
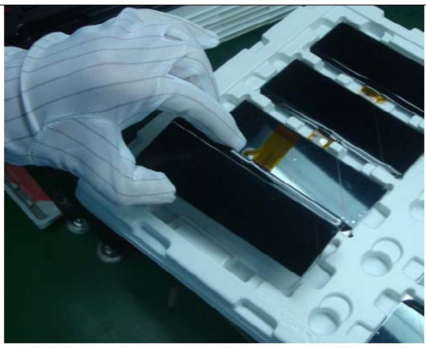
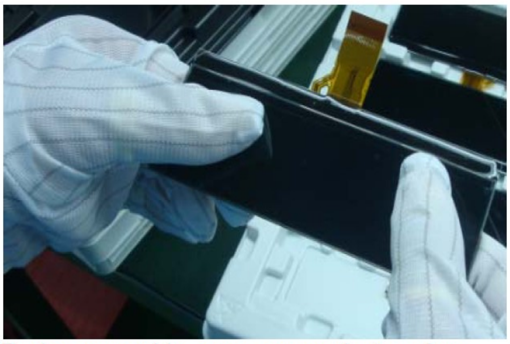
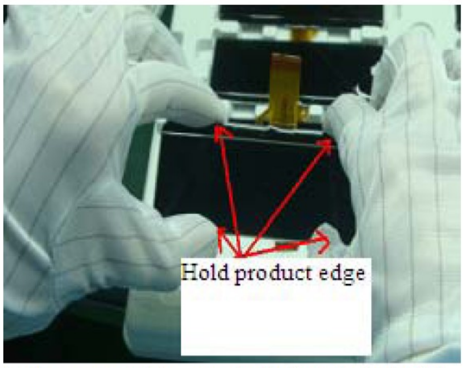
9. Packing Specification

TBD

10. Handling Cautions

10.1 Packing Removal and Handling Requirement

Requirement	Wrong	Correct
Get one package each times & hold the package by both hands with proper ESD shielding	 Hold the modules by one hand and without proper ESD shielding (Fail)	 Hold the modules by both hands (Pass)
Prohibit to stack inner package over 3 layers	 Over 3 layers (Fail)	 Not exceed 3 layers (Pass)
Total packing tray height must within 40 cm	 Over 40 cm (Fail)	 Lower than 40 cm (Pass)

Requirement	Wrong	Correct
Packing tray must rotate 180° in each layer when stack together	 Tray without 180° rotation, It will have pressure on the module (Fail)	 Tray with 180° rotation (Pass)
Prohibit to touch product surface by fingers	 Hold product and touch its surface (Fail)	 Hold product edge by hand (Pass)
During assembly, prohibit to press on product surface by fingers, Must hold the product edges by both hands	 During assembly, press on product surface (Fail)	 During assembly, use both hands to hold Product edge only (Pass)

10.2 Mounting of module

- Please power off the display module before it is disconnected or connected to the application.
- If the connection to the application is not good, following problems may result.
    1. Significant noise on signals between display module and application
    2. Unstable display performance
    3. Parts on the module will be heat up or damaged
- The LCD module must be handled with care.
- Protective film (Laminator) is applied on surface for protection against scratches and dirts. Please avoid electrostatic charge build-up when peeling off the laminator.

10.3 Precautions in Mounting

- When metal part of the LCD module (shielding lid and rear case) is soiled, wipe it with soft dry cloth.
- Wipe off water drops or finger grease immediately when found. Prolonged contact with water may cause discoloration or spots.
- The LCD module contains glass which breaks or cracks easily if dropped or bumped on hard surface. Please handle with care.
- The display and IC on module are sensitive to electrostatic discharge; please make sure equipments and operators are properly ground before and during handling.

10.4 Adjusting module

- Adjusting volumes on the rear face of the module have been set to its optimal before shipment. Therefore, do not change any adjusted values. If adjusted values are changed, the display may not perform to specification.

10.5 Others

- Do not expose the module to direct sunlight or intensive ultraviolet rays for prolonged hours
- Store the module at room temperature condition.
- If LCD panel breaks, liquid crystal may escape from the panel. Avoid bringing it to eyes or mouth contact. When liquid crystal sticks on hands, clothes or feet, wash it out immediately with soap.
- Observe all other precautionary requirements as in handling general electronic components.
- Please adjust the voltage of common electrode as materials of attachment by 1 module.
- Do not expose the display module to harmful gases such as acid and alkali gasses, which will corrode electronic components.
- Do not disassemble the display module because it can cause permanent damage and will void the warranty agreement.

11. Definitions

Data sheet status	
Objective Specification	This data sheet contains target or goal specifications for product development.
Preliminary Specification	This data sheet contains preliminary data; supplementary data may be published later.
Product Specification	This data sheet contains final product specification.
Limiting values	
Limiting values given are in accordance with the Absolute Maximum Rating. Stress above one or more of the limiting values may cause permanent damage to the device. These are stress ratings only and operating of the device at these or any other conditions above those given in the Characteristics sections of the specification is not implied. Expose to limiting values for extended periods may affect device reliability. Device is functional within the limiting conditions doesn't imply the same performance over the covered conditions, customer is required to decide the best range for the final applications.	

12. Life Support Applications

These products are not designed for use in life saving appliances, devices or systems where malfunctioning of these products can reasonably be expected to result in personal injury. Customers using or selling these products for use in such applications do so at their own risk and agree full non liability of Varitronix Limited for any damages or losses resulting from such improper use or sale.

13. Appendix